- Chi-square test,
- ANOVA
- Regression Analysis
- Time Series Analysis
- Multivariate Analysis

Hypothesis Testing for Variance and Standard Deviation

The **χ^2-test for a variance or standard deviation** is a statistical test for a population variance or standard deviation. The χ^2-test can be used when the population is normal.

The **test statistic** is s^2 and the **standardized test statistic**

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

follows a chi-square distribution with degrees of freedom d.f. = $n - 1$.

Using the χ^2-Test for a Variance or Standard Deviation

	In Words	*In Symbols*
1.	State the claim mathematically and verbally. Identify the null and alternative hypotheses.	State H_0 and H_a.
2.	Specify the level of significance.	Identify α.
3.	Determine the degrees of freedom and sketch the sampling distribution.	d.f. $= n - 1$
4.	Determine any critical values.	Use Table 6 in Appendix B.

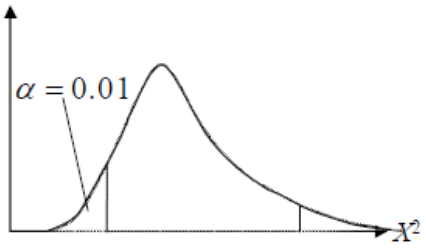
Using the χ2-Test for a Variance or Standard Deviation

In Words	*In Symbols*
5. Determine any rejection regions.	
6. Find the standardized test statistic.	$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$
7. Make a decision to reject or fail to reject the null hypothesis.	If χ^2 is in the rejection region, reject H_0. Otherwise, fail to reject H_0.
8. Interpret the decision in the context of the original claim.	

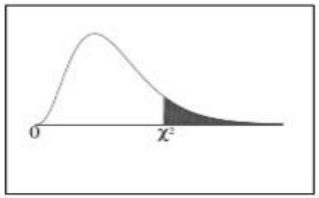
A college professor claims that the standard deviation for students taking a statistics test is less than 30. 10 tests are randomly selected and the standard deviation is found to be 28.8 Test this professor's claim at the $\alpha = 0.01$ level.

H_0: $\sigma \geq 30$ $\quad\quad$ H_a: $\sigma < 30$ (Claim)

This is a left-tailed test with d.f.= 9 and $\alpha = 0.01$.



Chi-Square Distribution Table



The shaded area is equal to α for $\chi^2 = \chi^2_\alpha$.

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

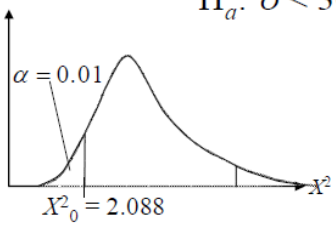
Example continued:

A college professor claims that the standard deviation for students taking a statistics test is less than 30. 10 tests are randomly selected and the standard deviation is found to be 28.8. Test this professor's claim at the $\alpha = 0.01$ level.

H_0: $\sigma \geq 30$

H_a: $\sigma < 30$ (Claim)

$\chi^2_0 = 2.088$



$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(10-1)(28.8)^2}{30^2} \approx 8.29$$

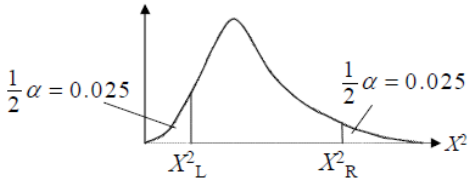
Fail to reject H_0.

At the 1% level of significance, there is not enough evidence to support the professor's claim.

Example:
A local balloon company claims that the variance for the time its helium balloons will stay afloat is 5 hours. A disgruntled customer wants to test this claim. She randomly selects 23 customers and finds that the variance of the sample is 4.5 seconds. At $\alpha = 0.05$, does she have enough evidence to reject the company's claim?

H_0: $\sigma^2 = 5$ (Claim) $\quad\quad$ H_a: $\sigma^2 \neq 5$

This is a two-tailed test with d.f.= 22 and $\alpha = 0.05$.



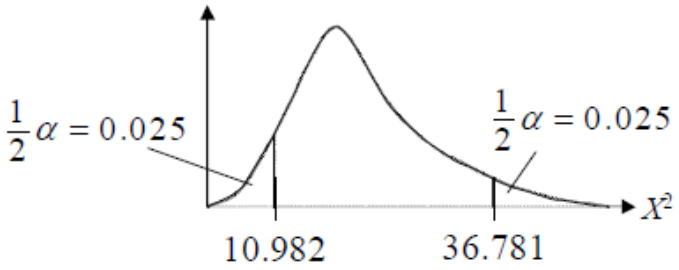
Continued.

Example continued:

A local balloon company claims that the variance for the time its helium balloons will stay afloat is 5 hours. A disgruntled customer wants to test this claim. She randomly selects 23 customers and finds that the variance of the sample is 4.5 seconds. At $\alpha = 0.05$, does she have enough evidence to reject the company's claim?

H_0: $\sigma^2 = 5$ (Claim) $\quad\quad$ H_a: $\sigma^2 \neq 5$

The critical values are $\chi^2_L = 10.982$ and $\chi^2_R = 36.781$.

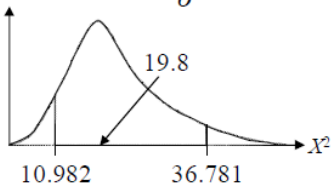


Continued.

Example continued:
A local balloon company claims that the variance for the time one of its helium balloons will stay afloat is 5 hours. A disgruntled customer wants to test this claim. She randomly selects 23 customers and finds that the variance of the sample is 4.5 seconds. At $\alpha = 0.05$, does she have enough evidence to reject the company's claim?

H_0: $\sigma^2 = 5$ (Claim) $\quad$ H_a: $\sigma^2 \neq 5$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(23-1)(4.5)}{5} = 19.8$$ Fail to reject H_0.



At $\alpha = 0.05$, there is not enough evidence to reject the claim that the variance of the float time is 5 hours.

Which Test?

Outcome Variable	Are the observations correlated?	
	independent	correlated
Binary or categorical (e.g. fracture, yes/no)	**Chi-square test:** compares proportions between two or more groups **Relative risks:** odds ratios or risk ratios **Logistic regression:** multivariate technique used when outcome is binary; gives multivariate-adjusted odds ratios	**McNemar's chi-square test:** compares binary outcome between correlated groups (e.g., before and after) **Conditional logistic regression:** multivariate regression technique for a binary outcome when groups are correlated (e.g., matched data) **GEE modeling:** multivariate regression technique for a binary outcome when groups are correlated (e.g., repeated measures)

Comparing more than two groups…

Outcome Variable	Are the observations independent or correlated?		Alternatives if the normality assumption is violated (and small sample size):
	independent	correlated	
Continuous (e.g. pain scale, cognitive function)	**t-test:** compares means between two independent groups **ANOVA:** compares means between more than two independent groups **Pearson's correlation coefficient** (linear correlation): shows linear correlation between two continuous variables **Linear regression:** multivariate regression technique used when the outcome is continuous; gives slopes	**Paired ttest:** compares means between two related groups (e.g., the same subjects before and after) **Repeated-measures ANOVA:** compares changes over time in the means of two or more groups (repeated measurements) **Mixed models/GEE modeling**: multivariate regression techniques to compare changes over time between two or more groups; gives rate of change over time	Non-parametric statistics **Wilcoxon sign-rank test**: non-parametric alternative to the paired ttest **Wilcoxon sum-rank test** (=Mann-Whitney U test): non parametric alternative to the ttest **Kruskal-Wallis test:** non-parametric alternative to ANOVA **Spearman rank correlatio coefficient:** non-parametric alternative to Pearson's correlatio coefficient

- if there is a significant association between categorical variables
- Types of chi-square tests:
  - **Test of Independence**
    - ✓ The **chi-square test of independence** assesses whether observed frequencies are dependent on (i.e., contingent on) certain conditions.
    - ✓ Determines whether there is a significant **association between two categorical variables**.
    - ✓ Testing if there is a relationship between gender (male/female) and preference for a product (like/dislike).
  - **Goodness-of-Fit Test**
    - ✓ The **chi-square goodness of fit test** is used to compare the observed frequencies in a data sample with the frequencies based on some prior expectation – either empirical or theoretical
    - ✓ Determines whether the **observed frequency distribution of a categorical variable matches an expected distribution**.
    - ✓ Testing if the distribution of blood types in a population matches the expected distribution (e.g., 40% O, 30% A, 20% B, 10% AB).

Example

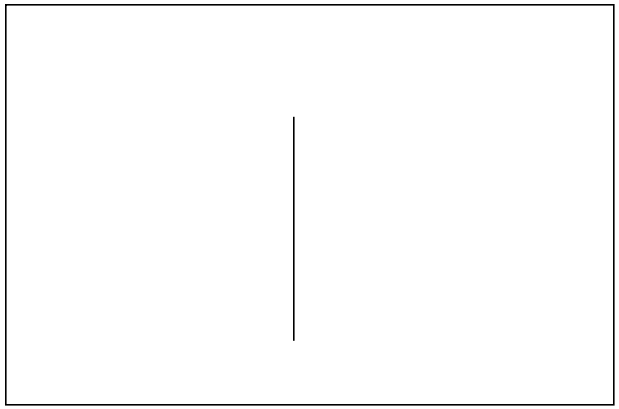
- https://www.youtube.com/watch?v=zOvUQWOzTlc

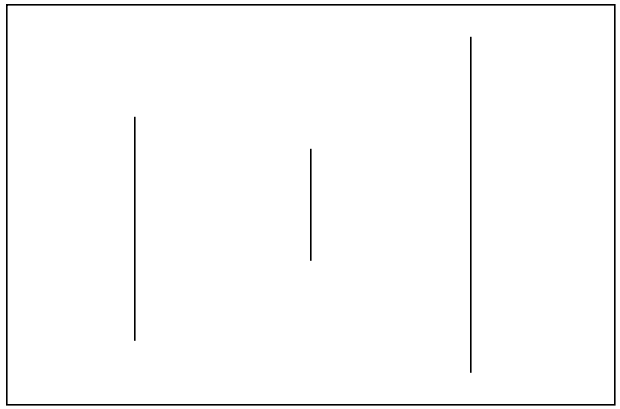
- Asch, S.E. (1955). Opinions and social pressure. *Scientific American*, 193, 31-35.
- https://www.youtube.com/watch?v=WkK5eA_qhFk

The Experiment

- A Subject volunteers to participate in a “visual perception study.”
- Everyone else in the room is actually a conspirator in the study (unbeknownst to the Subject).
- The “experimenter” reveals a pair of cards…

The Task Cards





The Experiment

- Everyone goes around the room and says which comparison line (A, B, or C) is correct; the true Subject always answers last – after hearing all the others' answers.
- The first few times, the 7 "conspirators" give the correct answer.
- Then, they start purposely giving the (obviously) wrong answer.
- 75% of Subjects tested went along with the group's consensus at least once.

Further Results

- In a further experiment, group size (number of conspirators) was altered from 2-10.

- ***Does the group size alter the proportion of subjects who conform?***

The Chi-Square test

Conformed?	Number of group members?				
	2	4	6	8	10
Yes	20	50	75	60	30
No	80	50	25	40	70

- 20 + 50 + 75 + 60 + 30 = 235 conformed
- out of 500 experiments.

- Overall likelihood of conforming = 235/500 = .47

Calculating the expected, in general

- Null hypothesis: variables are independent
- Recall that under independence:

  P(A)*P(B)=P(A&B)
- Therefore, calculate the marginal probability of B and the marginal probability of A. Multiply P(A)*P(B)*N to get the expected cell count.

Expected frequencies if no association between group size and conformity…

Conformed?	Number of group members?				
	2	4	6	8	10
Yes	47	47	47	47	47
No	53	53	53	53	53

- Do observed and expected differ more than expected due to chance?

Chi-Square test

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

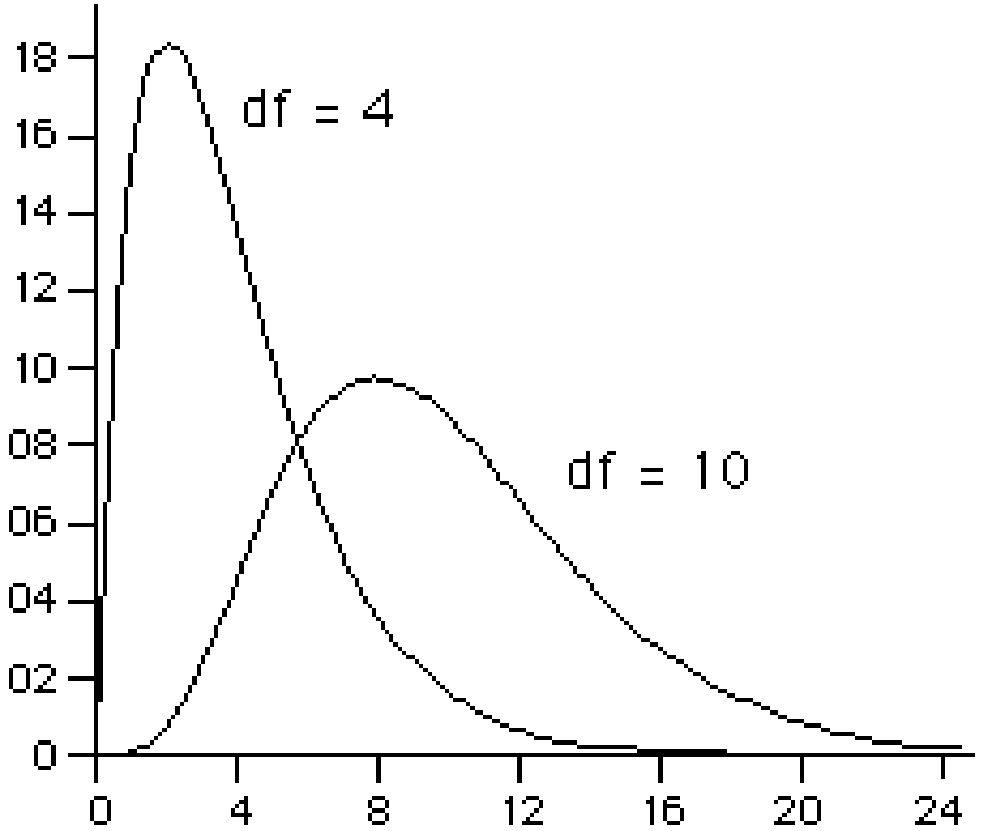
$$\chi_4^2 = \frac{(20-47)^2}{47} + \frac{(50-47)^2}{47} + \frac{(75-47)^2}{47} + \frac{(60-47)^2}{47} + \frac{(30-47)^2}{47} +$$
$$\frac{(80-53)^2}{53} + \frac{(50-53)^2}{53} + \frac{(25-53)^2}{53} + \frac{(40-53)^2}{53} + \frac{(70-53)^2}{53} \approx 85$$

Degrees of freedom = (rows-1)*(columns-1)=(2-1)*(5-1)=4

The Chi-Square distribution:

is sum of squared normal deviates

$$\chi^2_{df} = \sum_{i=1}^{df} Z^2; \text{ where } Z \sim \text{Normal}(0,1)$$



The expected value and variance of a chi-square:

$E(x)=df$

$Var(x)=2(df)$

Chi-Square test

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

$$\chi_4^2 = \frac{(20-47)^2}{47} + \frac{(50-47)^2}{47} + \frac{(75-47)^2}{47} + \frac{(60-47)^2}{47} + \frac{(30-47)^2}{47} +$$
$$\frac{(80-53)^2}{53} + \frac{(50-53)^2}{53} + \frac{(25-53)^2}{53} + \frac{(40-53)^2}{53} + \frac{(70-53)^2}{53} \approx 85$$

Degrees of freedom = (rows-1)*(columns-1)=(2-1)*(5-1)=4

Rule of thumb: if the chi-square statistic is much greater than it's degrees of freedom, indicates statistical significance. Here 85>>4.

Binary or categorical outcomes (proportions)

Outcome Variable	Are the observations correlated?		Alternative to the chi-square test if sparse cells:
	independent	correlated	
Binary or categorical (e.g. fracture, yes/no)	**Chi-square test:** compares proportions between two or more groups **Relative risks:** odds ratios or risk ratios **Logistic regression:** multivariate technique used when outcome is binary; gives multivariate-adjusted odds ratios	**McNemar's chi-square test:** compares binary outcome between correlated groups (e.g., before and after) **Conditional logistic regression:** multivariate regression technique for a binary outcome when groups are correlated (e.g., matched data) **GEE modeling:** multivariate regression technique for a binary outcome when groups are correlated (e.g., repeated measures)	**Fisher's exact test:** compares proportions between independent groups when there are sparse data (np <5). **McNemar's exact test:** compares proportions between correlated groups when there are sparse data (np <5).

ANOVA
(**AN**alysis **O**f **VA**riance)

- ***Idea***: For two or more groups, test difference between means, for quantitative normally distributed variables.
- Just an extension of the t-test (an ANOVA with only two groups is mathematically equivalent to a t-test).

One-Way Analysis of Variance

- Assumptions, same as ttest
  - Normally distributed outcome
  - Equal variances between the groups
  - Groups are independent

Hypotheses of One-Way ANOVA

$$\mathbf{H_0 : \mu_1 = \mu_2 = \mu_3 = \cdots}$$

$$\boldsymbol{H_1}\textbf{ : Not all of the population means are the same}$$

ANOVA

- It's like this: If I have three groups to compare:
  - I could do three pair-wise t-tests, but this would increase my type I error
  - So, instead I want to look at the pairwise differences "all at once."
  - To do this, I can recognize that variance is a statistic that let's me look at more than one difference at a time…

The "F-test"

Is the difference in the means of the groups more than background noise (=variability within groups)?

Summarizes the mean differences between all groups at once.

$$F = \frac{Variability\ between\ groups}{Variability\ within\ groups}$$

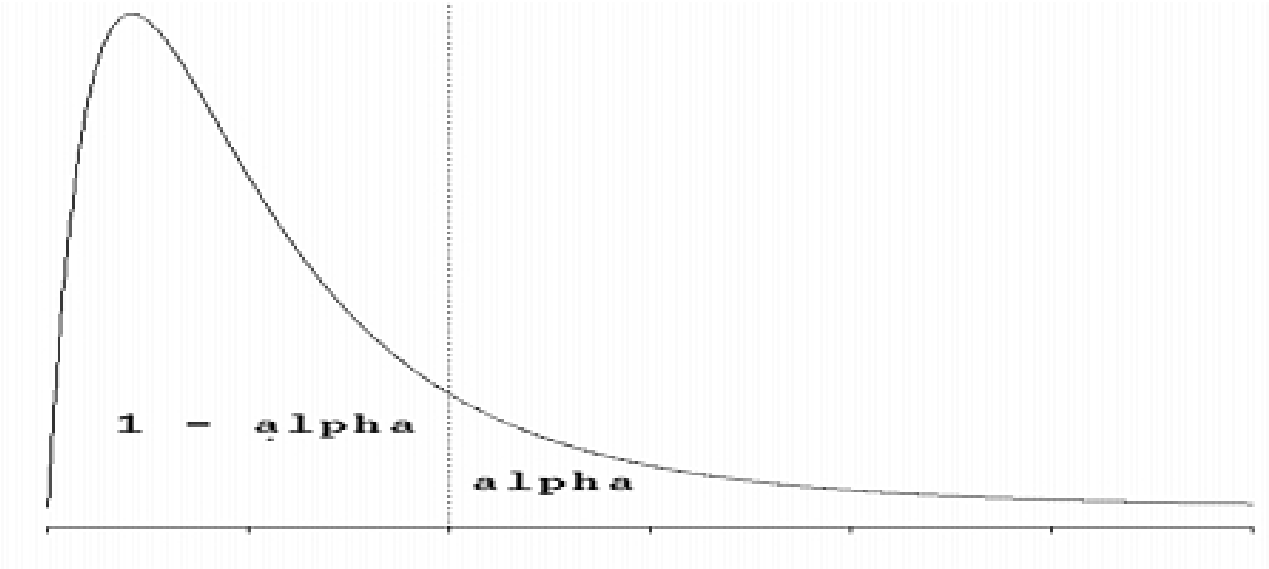
Analogous to pooled variance from a t-test.

Recall, we have already used an "F-test" to check for equality of variances→ If F>>1 (indicating unequal variances), use unpooled variance in a t-test.

The F-distribution

- The F-distribution is a continuous probability distribution that depends on two parameters n and m (numerator and denominator degrees of freedom, respectively):

http://www.econtools.com/jevons/java/Graphics2D/FDist.html



The F-distribution

- A ratio of variances follows an F-distribution:

$$\frac{\sigma^2_{between}}{\sigma^2_{within}} \sim F_{n,m}$$

- The F-test tests the hypothesis that two variances are equal.
- F will be close to 1 if sample variances are equal.

$$H_0 : \sigma^2_{between} = \sigma^2_{within}$$

$$H_a : \sigma^2_{between} \neq \sigma^2_{within}$$

How to calculate ANOVA's by hand…

Treatment 1	Treatment 2	Treatment 3	Treatment 4
y_{11}	y_{21}	y_{31}	y_{41}
y_{12}	y_{22}	y_{32}	y_{42}
y_{13}	y_{23}	y_{33}	y_{43}
y_{14}	y_{24}	y_{34}	y_{44}
y_{15}	y_{25}	y_{35}	y_{45}
y_{16}	y_{26}	y_{36}	y_{46}
y_{17}	y_{27}	y_{37}	y_{47}
y_{18}	y_{28}	y_{38}	y_{48}
y_{19}	y_{29}	y_{39}	y_{49}
y_{110}	y_{210}	y_{310}	y_{410}

n=10 obs./group

k=4 groups

$$\bar{y}_{1\bullet} = \frac{\sum_{j=1}^{10} y_{1j}}{10} \qquad \bar{y}_{2\bullet} = \frac{\sum_{j=1}^{10} y_{2j}}{10} \qquad \bar{y}_{3\bullet} = \frac{\sum_{j=1}^{10} y_{3j}}{10} \qquad \bar{y}_{4\bullet} = \frac{\sum_{j=1}^{10} y_{4j}}{10}$$

The group means

$$\frac{\sum_{j=1}^{10} (y_{1j} - \bar{y}_{1\bullet})^2}{10-1} \qquad \frac{\sum_{j=1}^{10} (y_{2j} - \bar{y}_{2\bullet})^2}{10-1} \qquad \frac{\sum_{j=1}^{10} (y_{3j} - \bar{y}_{3\bullet})^2}{10-1} \qquad \frac{\sum_{j=1}^{10} (y_{4j} - \bar{y}_{4\bullet})^2}{10-1}$$

The (within) group variances

Sum of Squares Within (SSW), or Sum of Squares Error (SSE)

$$\frac{\sum_{j=1}^{10}(y_{1j}-\bar{y}_{1\bullet})^2}{10-1} \quad \frac{\sum_{j=1}^{10}(y_{2j}-\bar{y}_{2\bullet})^2}{10-1} \quad \frac{\sum_{j=1}^{10}(y_{3j}-\bar{y}_{3\bullet})^2}{10-1} \quad \frac{\sum_{j=1}^{10}(y_{4j}-\bar{y}_{4\bullet})^2}{10-1}$$

The (within) group variances

$$\sum_{j=1}^{10}(y_{1j}-\bar{y}_{1\bullet})^2 + \sum_{j=1}^{10}(y_{2j}-\bar{y}_{2\bullet})^2 + \sum_{j=3}^{10}(y_{3j}-\bar{y}_{3\bullet})^2 + \sum_{j=1}^{10}(y_{4j}-\bar{y}_{4\bullet})^2$$

$$= \sum_{i=1}^{4}\sum_{j=1}^{10}(y_{ij}-\bar{y}_{i\bullet})^2$$

Sum of Squares Within (SSW) (or SSE, for chance error)

Sum of Squares Between (SSB), or Sum of Squares Regression (SSR)

Overall mean of all 40 observations ("grand mean")

$$\bar{\bar{y}}_{\bullet\bullet} = \frac{\sum_{i=1}^{4}\sum_{j=1}^{10} y_{ij}}{40}$$

$$10x\sum_{i=1}^{4}(\bar{y}_{i\bullet} - \bar{\bar{y}}_{\bullet\bullet})^2 \longleftarrow$$

Sum of Squares Between (SSB). Variability of the group means compared to the grand mean (the variability due to the treatment).

Total Sum of Squares (SST)

$$\sum_{i=1}^{4}\sum_{j=1}^{10}(y_{ij}-\bar{\bar{y}}_{\bullet\bullet})^2$$

Total sum of squares(TSS). Squared difference of every observation from the overall mean.

Partitioning of Variance

$$\sum_{i=1}^{4}\sum_{j=1}^{10}(y_{ij}-\bar{y}_{i\bullet})^2 \; +10\times\sum_{i=1}^{4}(\bar{y}_{i\bullet}-\bar{\bar{y}}_{\bullet\bullet})^2 \; = \; \sum_{i=1}^{4}\sum_{j=1}^{10}(y_{ij}-\bar{\bar{y}}_{\bullet\bullet})^2$$

SSW + SSB = TSS

ANOVA Table

Source of variation	d.f.	Sum of squares	Mean Sum of Squares	F-statistic	p-value
Between (k groups)	k-1	SSB (sum of squared deviations of group means from grand mean)	SSB/k-1	$\frac{SSB/k-1}{SSW/nk-k}$	Go to $F_{k-1,nk-k}$ chart
Within (n individuals per group)	nk-k	SSW (sum of squared deviations of observations from their group mean)	s^2=SSW/nk-k		
Total variation	nk-1	TSS (sum of squared deviations of observations from grand mean)		TSS=SSB + SSW	

ANOVA=t-test

$$SSB = n\sum_{i=1}^{n}(\bar{X}_n - (\frac{\bar{X}_n + \bar{Y}_n}{2}))^2 + n\sum_{i=1}^{n}(\bar{Y}_n - (\frac{\bar{X}_n + \bar{Y}_n}{2}))^2 =$$

$$n\sum_{i=1}^{n}(\frac{\bar{X}_n}{2} - \frac{\bar{Y}_n}{2})^2 + n\sum_{i=1}^{n}(\frac{\bar{Y}_n}{2} - \frac{\bar{X}_n}{2})^2$$

$$n((\frac{\bar{X}_n}{2})^2 + (\frac{\bar{Y}_n}{2})^2 - 2\frac{\bar{X}_n * \bar{Y}_n}{2} + (\frac{\bar{Y}_n}{2})^2 + (\frac{\bar{X}_n}{2})^2 - 2\frac{\bar{X}_n * \bar{Y}_n}{2}) =$$

$$n(\bar{X}_n{}^2 - 2\bar{X}_n * \bar{Y}_n + \bar{Y}_n{}^2) = n(\bar{X}_n - \bar{Y}_n)^2$$

Source of variation	d.f.	Sum of squares	Mean Sum of Squares	F-statistic	p-value
Between (2 groups)	1	SSB (squared difference in means multiplied by n)	Squared difference in means times n	$\frac{n(\bar{X}-\bar{Y})^2}{s_p{}^2} = (\frac{(\bar{X}-\bar{Y})}{\sqrt{\frac{s_p{}^2}{n} + \frac{s_p{}^2}{n}}})^2 = (t_{2n-2})^2$	Go to $F_{1,\,2n-2}$ Chart→ notice values are just $(t_{2n-2})^2$
Within	2n-2	SSW equivalent to numerator of pooled variance	Pooled variance		
Total variation	2n-1	TSS			

Example

Treatment 1	Treatment 2	Treatment 3	Treatment 4
60 inches	50	48	47
67	52	49	67
42	43	50	54
67	67	55	67
56	67	56	68
62	59	61	65
64	67	61	65
59	64	60	56
72	63	59	60
71	65	64	65

Example

Step 1) calculate the sum of squares between groups:

Treatment 1	Treatment 2	Treatment 3	Treatment 4
60 inches	50	48	47
67	52	49	67
42	43	50	54
67	67	55	67
56	67	56	68
62	59	61	65
64	67	61	65
59	64	60	56
72	63	59	60
71	65	64	65

Mean for group 1 = 62.0

Mean for group 2 = 59.7

Mean for group 3 = 56.3

Mean for group 4 = 61.4

Grand mean= 59.85

$SSB = [(62-59.85)^2 + (59.7-59.85)^2 + (56.3-59.85)^2 + (61.4-59.85)^2]\ xn\ per\ group = 19.65x10 = 196.5$

Example

Step 2) calculate the sum of squares within groups:

$(60-62)^2+(67-62)^2+(42-62)^2+(67-62)^2+(56-62)^2+(62-62)^2+(64-62)^2+(59-62)^2+(72-62)^2+(71-62)^2+(50-59.7)^2+(52-59.7)^2+(43-59.7)^2+67-59.7)^2+(67-59.7)^2+(69-59.7)^2$…+….(sum of 40 squared deviations) = 2060.6

Treatment 1	Treatment 2	Treatment 3	Treatment 4
60 inches	50	48	47
67	52	49	67
42	43	50	54
67	67	55	67
56	67	56	68
62	59	61	65
64	67	61	65
59	64	60	56
72	63	59	60
71	65	64	65

Step 3) Fill in the ANOVA table

Source of variation	d.f.	Sum of squares	Mean Sum of Squares	F-statistic	p-value
Between	3	196.5	65.5	1.14	.344
Within	36	2060.6	57.2		
Total	39	2257.1			

Step 3) Fill in the ANOVA table

Source of variation	d.f.	Sum of squares	Mean Sum of Squares	F-statistic	p-value
Between	3	196.5	65.5	1.14	.344
Within	36	2060.6	57.2	_	_
Total	39	2257.1	_	_	_

INTERPRETATION of ANOVA:

How much of the variance in height is explained by treatment group?

$R^{2=}$"Coefficient of Determination" = SSB/TSS = 196.5/2275.1=9%

Coefficient of Determination

$$R^2 = \frac{SSB}{SSB + SSE} = \frac{SSB}{SST}$$

The amount of variation in the outcome variable (dependent variable) that is explained by the predictor (independent variable).

Beyond one-way ANOVA

Often, you may want to test more than 1 treatment. ANOVA can accommodate more than 1 treatment or factor, so long as they are independent. Again, the variation partitions beautifully!

TSS = SSB1 + SSB2 + SSW

ANOVA example

Table 6. Mean micronutrient intake from the school lunch by school

		S1[a], *n*=25	S2[b], *n*=25	S3[c], *n*=25	*P*-value[d]
Calcium (mg)	Mean	117.8	158.7	206.5	0.000
	SD[e]	62.4	70.5	86.2	
Iron (mg)	Mean	2.0	2.0	2.0	0.854
	SD	0.6	0.6	0.6	
Folate (μg)	Mean	26.6	38.7	42.6	0.000
	SD	13.1	14.5	15.1	
Zinc (mg)	Mean	1.9	1.5	1.3	0.055
	SD	1.0	1.2	0.4	

[a] School 1 (most deprived; 40% subsidized lunches).
[b] School 2 (medium deprived; <10% subsidized).
[c] School 3 (least deprived; no subsidization, private school).
[d] ANOVA; significant differences are highlighted in bold ($P<0.05$).

FROM: Gould R, Russell J, Barker ME. School lunch menus and 11 to 12 year old children's food choice in three secondary schools in England-are the nutritional standards being met? *Appetite.* 2006 Jan;46(1):86-92.

Answer

Step 1) calculate the sum of squares between groups:

Mean for School 1 = 117.8

Mean for School 2 = 158.7

Mean for School 3 = 206.5

Grand mean: 161

SSB = $[(117.8-161)^2 + (158.7-161)^2 + (206.5-161)^2]$ *x25 per group*= 98,113

Answer

Step 2) calculate the sum of squares within groups:

S.D. for S1 = 62.4

S.D. for S2 = 70.5

S.D. for S3 = 86.2

Therefore, sum of squares within is:

$(24)[\ 62.4^2 + 70.5^2 + 86.2^2] = 391{,}066$

Answer

Step 3) Fill in your ANOVA table

Source of variation	d.f.	Sum of squares	Mean Sum of Squares	F-statistic	p-value
Between	2	98,113	49056	9	<.05
Within	72	391,066	5431		
Total	74	489,179			

**R^2=98113/489179=20%

School explains 20% of the variance in lunchtime calcium intake in these kids.

ANOVA summary

- A statistically significant ANOVA (F-test) only tells you that *at least* two of the groups differ, but not which ones differ.

- Determining *which* groups differ (when it's unclear) requires more sophisticated analyses to correct for the problem of multiple comparisons…

What is Regression?

Relationships

In science, we frequently measure two or more variables on the same individual (case, object, etc). We do this to explore the nature of the relationship among these variables. There are two basic types of relationships.

- Cause-and-effect relationships.
- Functional relationships.

Function: a mathematical relationship enabling us to predict what values of one variable (Y) correspond to given values of another variable (X).

- Y: is referred to as the *dependent variable*, the *response variable* or the *predicted variable*.
- X: is referred to as the *independent variable*, the *explanatory variable* or the *predictor variable*.

Examples

- The repair time for a computer
- The number of components which have to be changed

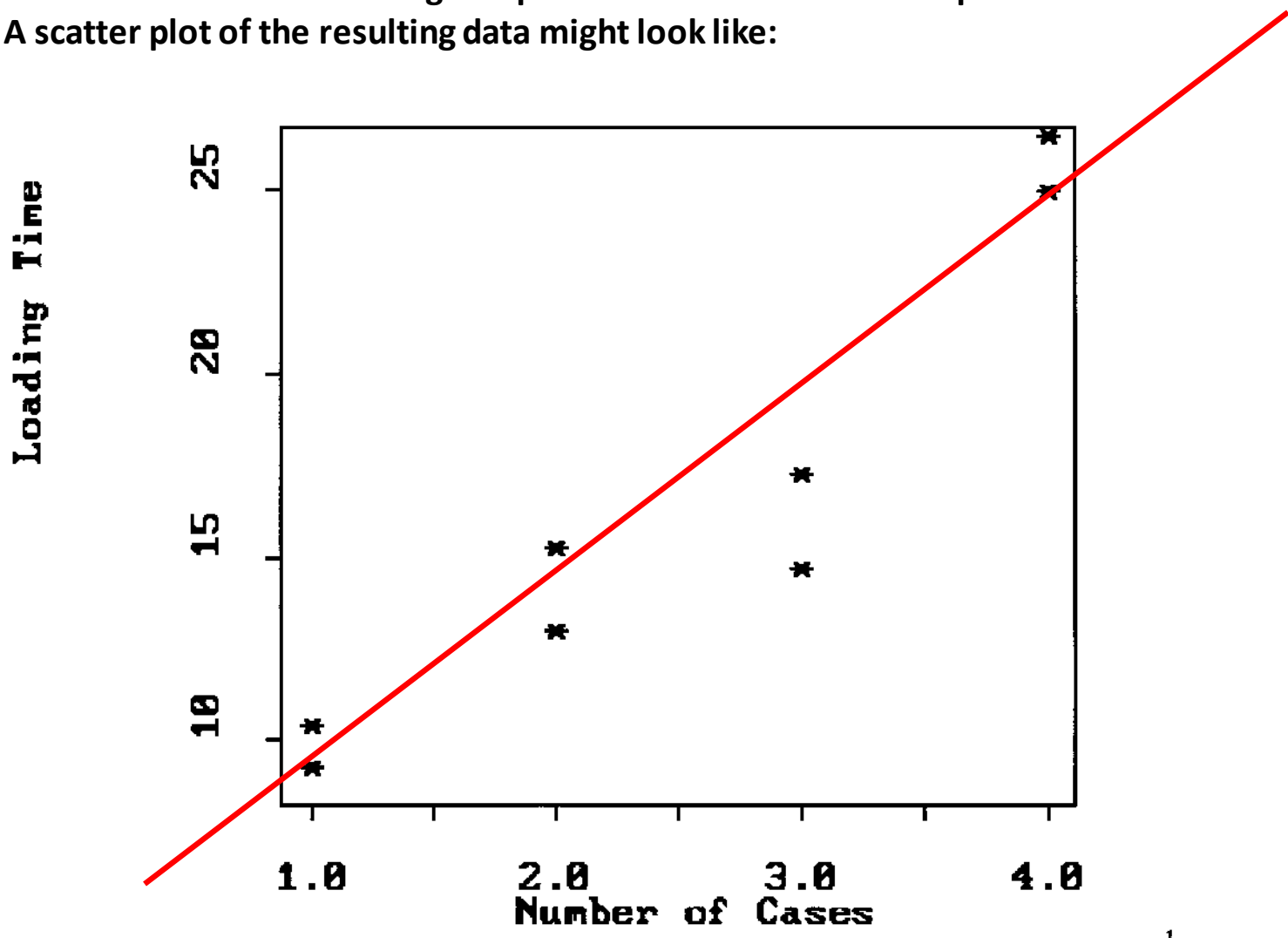
In each case, the statement can be read as; ***Y is a function of X***.

Two kinds of explanatory variables:

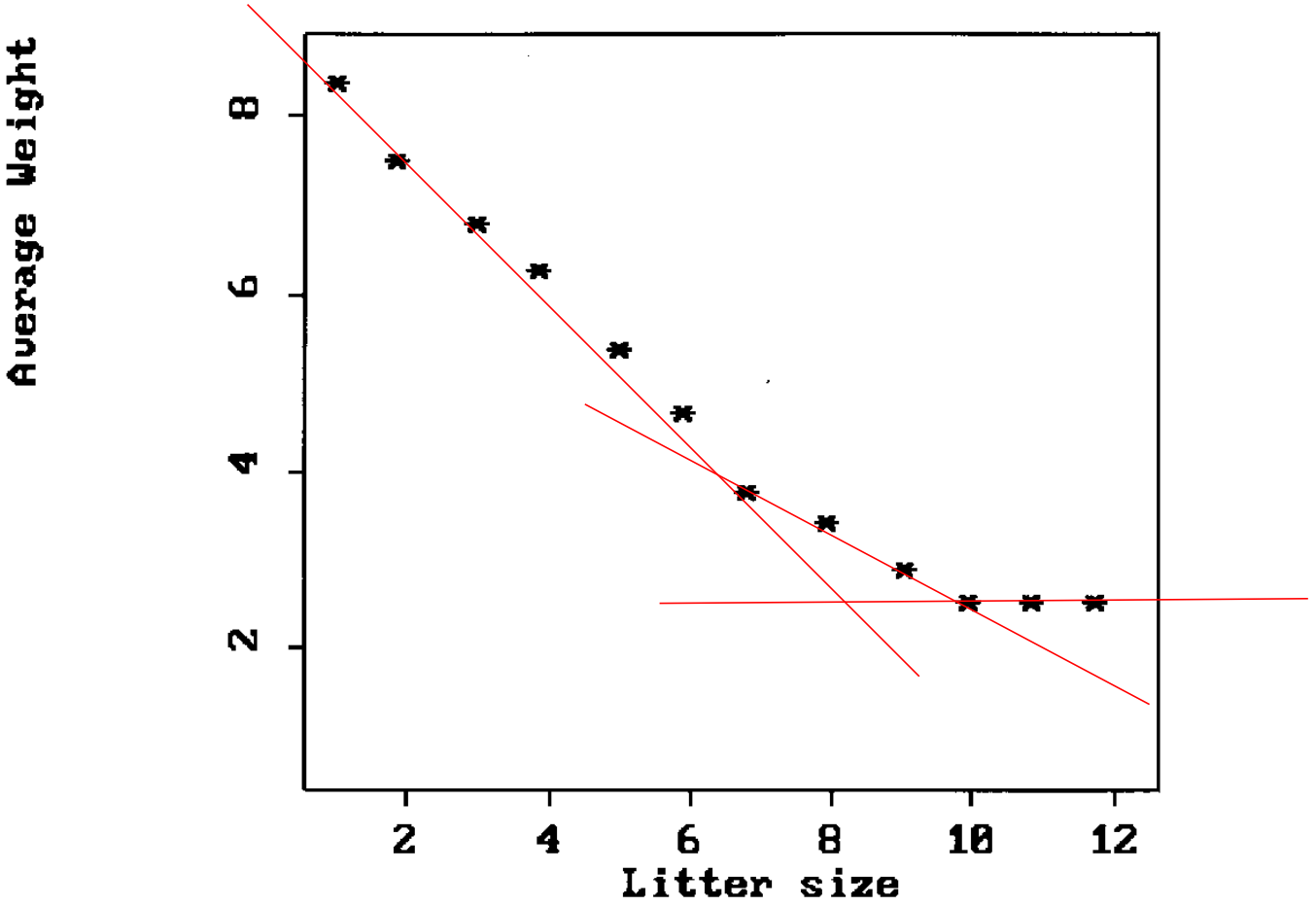
Those we can control

Those over which we have little or no control.

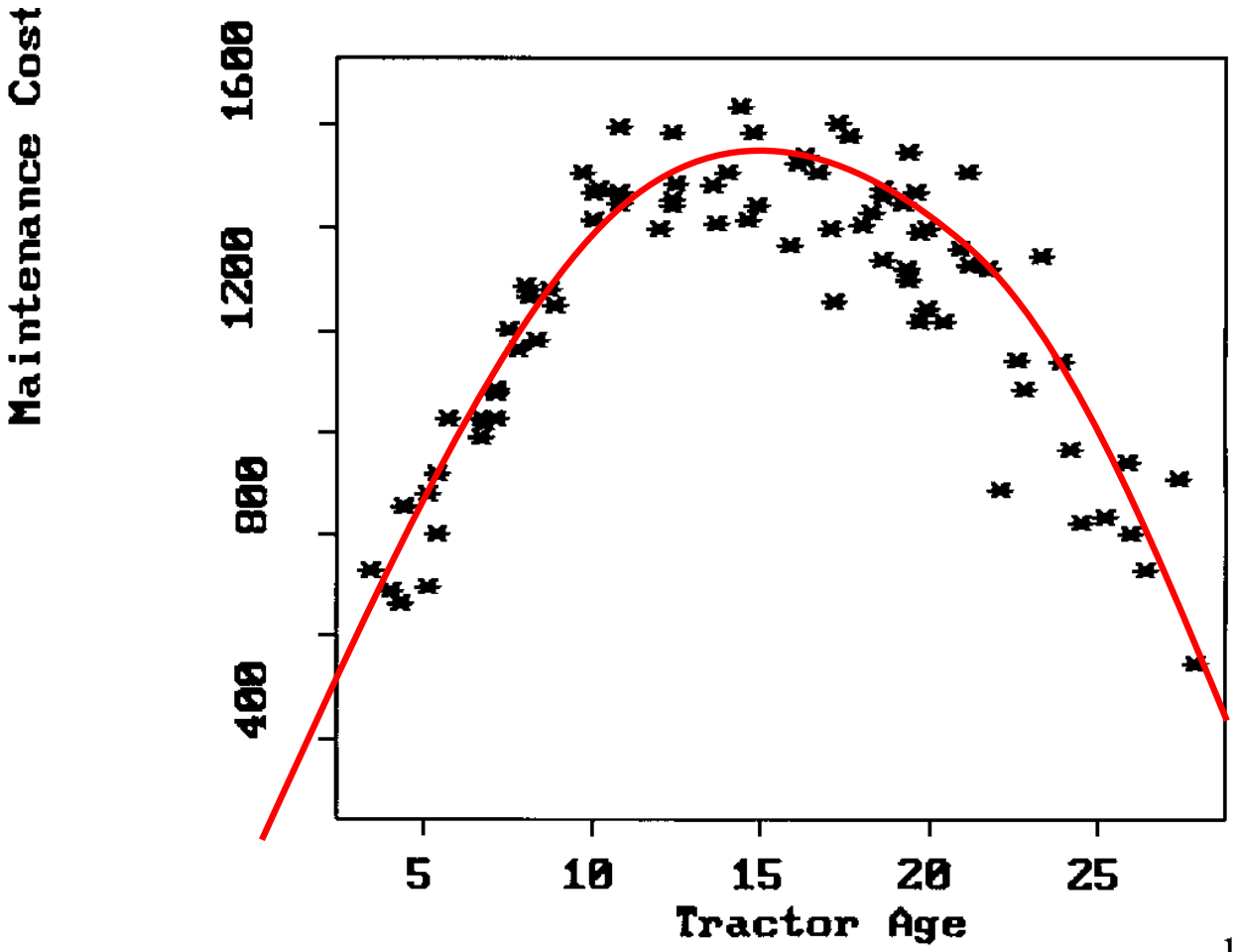
An operations supervisor measured how long it takes one of her drivers to put 1, 2, 3 and 4 cases of soft drink into a soft drink machine. In this case the levels of the explanatory variable, X are {1,2,3,4}, and she controls them. She might repeat the measurement a couple of times at each level of X. A scatter plot of the resulting data might look like:



A farm manager is interested in the relationship between litter size and average litter weight (average newborn piglet weight). She examines the farm records over the last couple of years and records the litter size and average weight for all births. A plot of the data pairs looks like the following:



A farm operations student is interested in the relationship between maintenance cost and age of farm tractors. He performs a telephone interview survey of the 52 commercial potato growers in Putnam County, FL. One part of the questionnaire provides information on tractor age and 1995 maintenance cost (fuel, lubricants, repairs, etc). A plot of these data might look like:



11.

Questions needing answers.

- What is the *association* between Y and X?
- How can changes in Y be explained by changes in X?
- What are the *functional relationships* between Y and X?

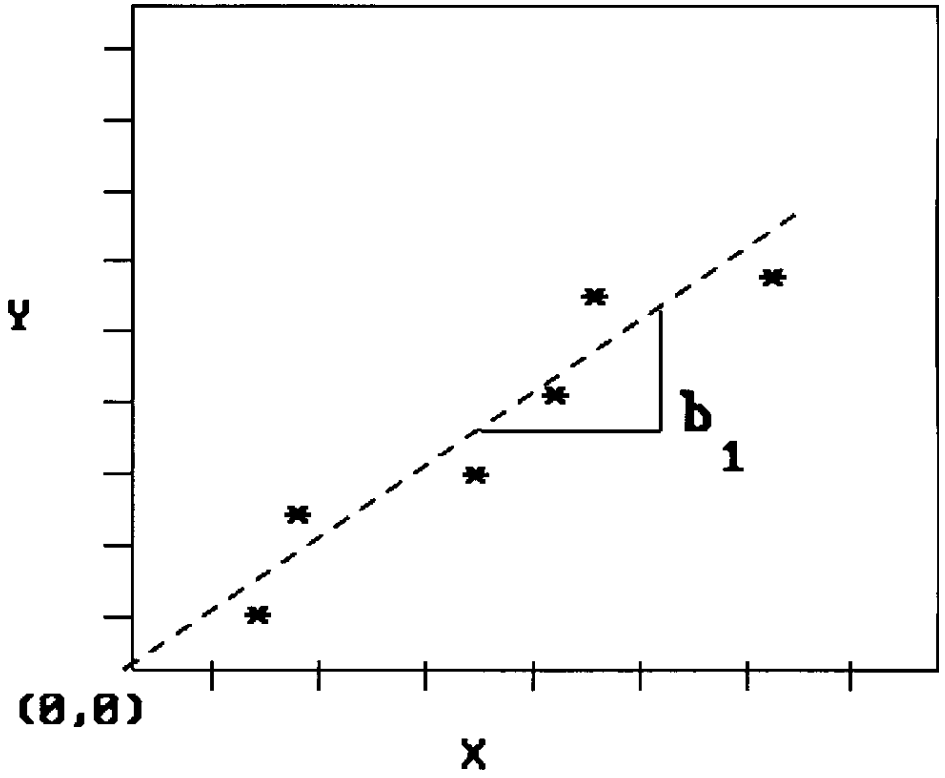
A functional relationship is symbolically written as:

Eq: 1 $$Y = f(X)$$

Example: A proportional relationship (e.g. fish weight to length).

$$Y = b_1 X$$

b_1 is the **slope** of the line.

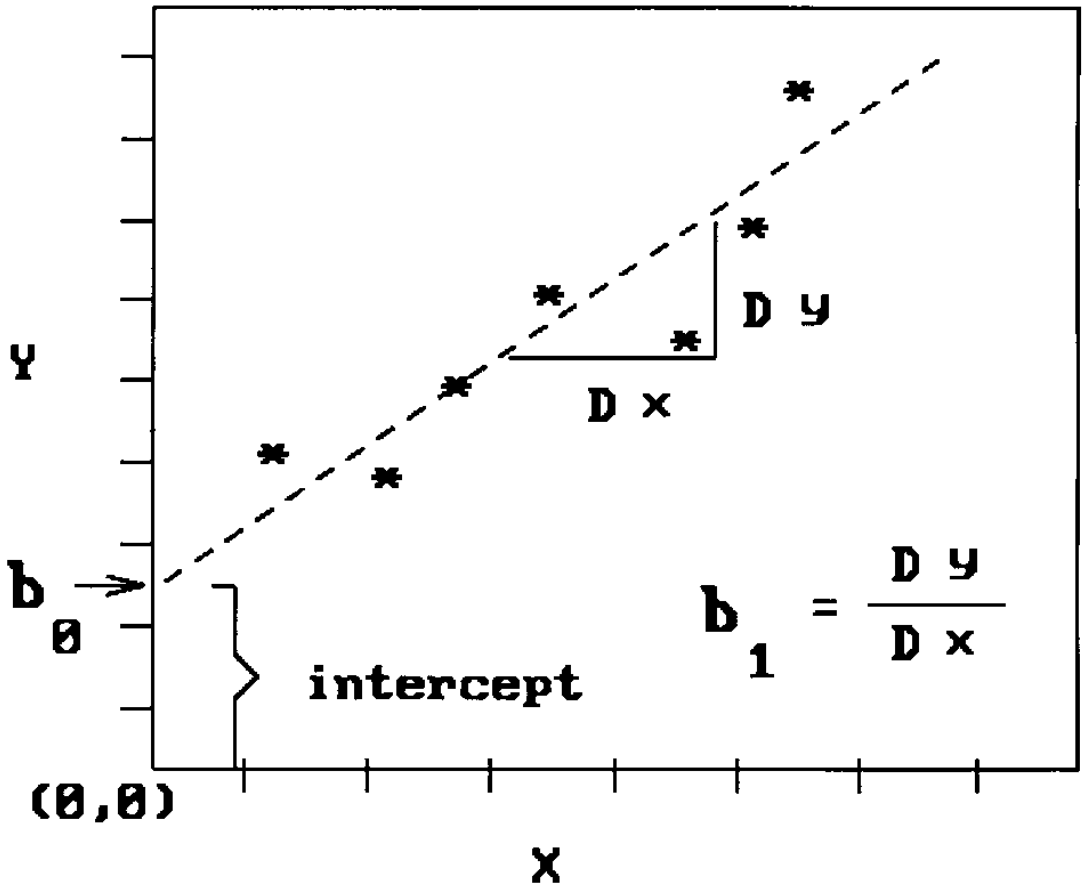


Example: Linear relationship (e.g. Y=cholesterol versus X=age)

$$Y = b_0 + b_1 X$$

b_0 is the intercept,

b_1 is the slope.



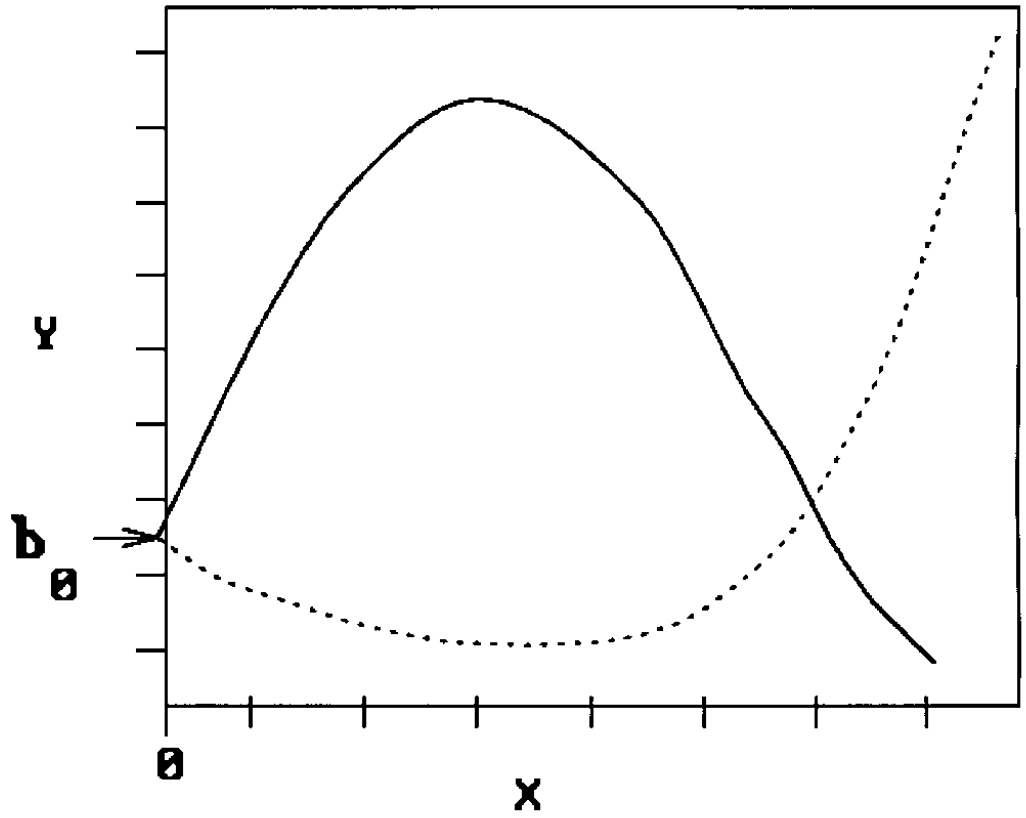
Example: Polynomial relationship (e.g. Y=crop yield vs. X=pH)

$$Y = b_0 + b_1 X + b_2 X^2$$

b_0: intercept,

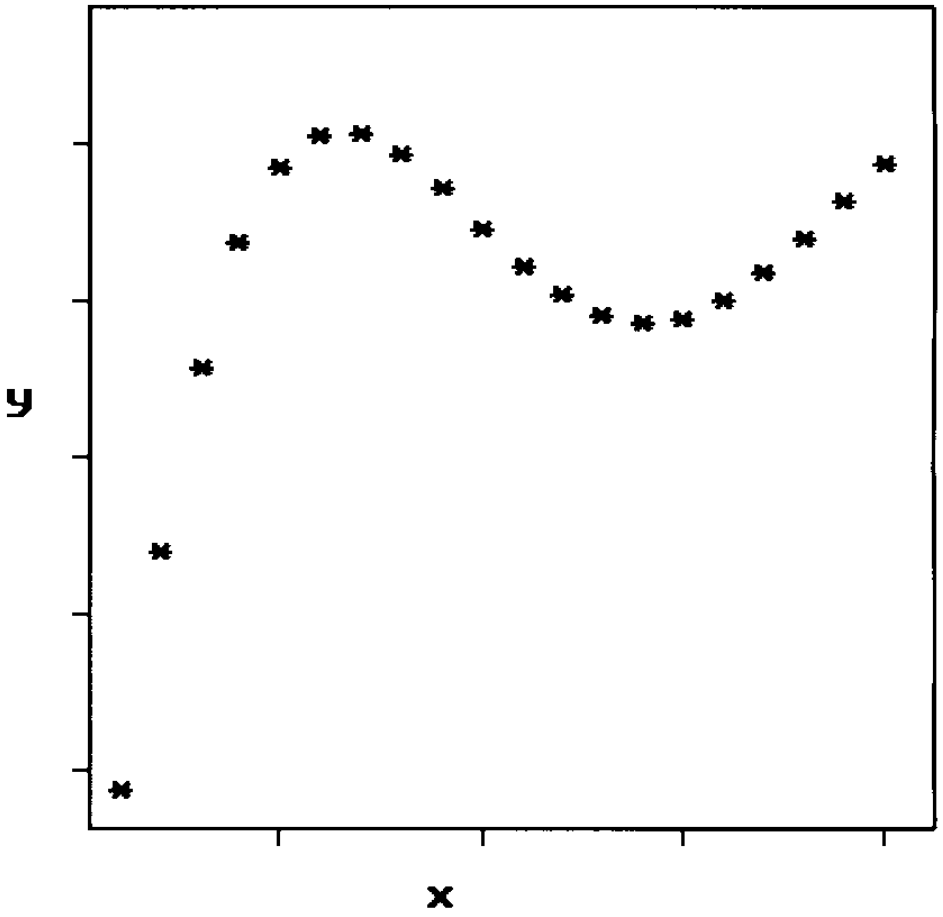
b_1: linear coefficient,

b_2: quadratic coefficient.



Nonlinear relationship:

$$Y = b_0 \sin(b_1 X + b_2 X^2)$$



Concerns:

- **The proposed functional relationship will not fit exactly, i.e. something is either wrong with the data (errors in measurement), or the model is inadequate (errors in specification).**
- **The relationship is not truly known until we assign values to the parameters of the model.**

The possibility of errors into the proposed relationship is acknowledged in the functional symbolism as follows:

Eq: 2 $$Y = f(X) + \varepsilon$$

ε is a random variable representing the result of both errors in model specification and measurement. As in AOV, the variance of ε is the **background variability** with respect to which we will assess the significance of the factors (explanatory variables).

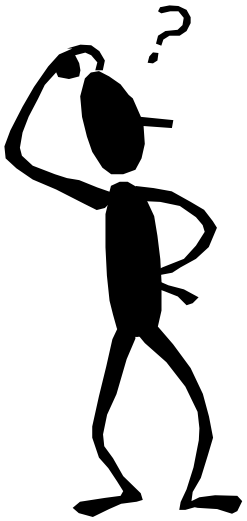
The error term:

Another way to emphasize

Eq: 3 $$\varepsilon = Y - f(X)$$

or, emphasizing that f(X) depends on unknown parameters.

Eq: 4 $$Y = f(X \mid \beta_0, \beta_1) + \varepsilon$$



What if we don't know the functional form of the relationship?

- Look at a scatter plot of the data for suggestions.
- Hypothesize about the nature of the underlying process. Often the hypothesized processes will suggest a functional form.

The straight line -- a conservative starting point.

Regression Analysis: the process of fitting a line to data.

Sir Francis Galton (1822-1911) -- a British anthropologist and meteorologist coined the term "regression".

Regression towards mediocrity in hereditary stature - the tendency of offspring to be smaller than large parents and larger than small parents. Referred to as "regression towards the mean".

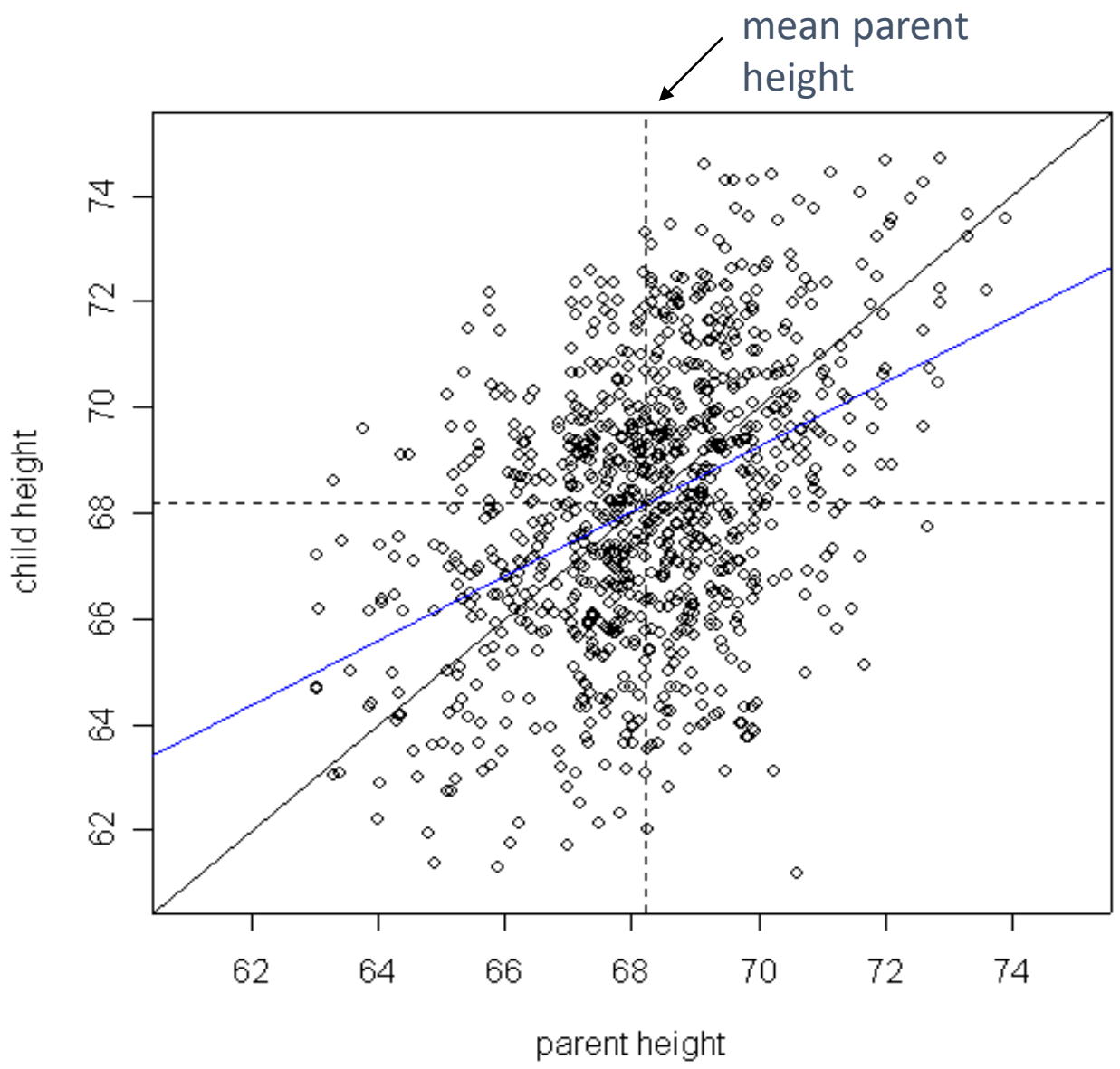
$$\hat{Y} = \bar{Y} + \frac{2}{3}(X - \bar{X})$$

Expected offspring height

Average sized offspring

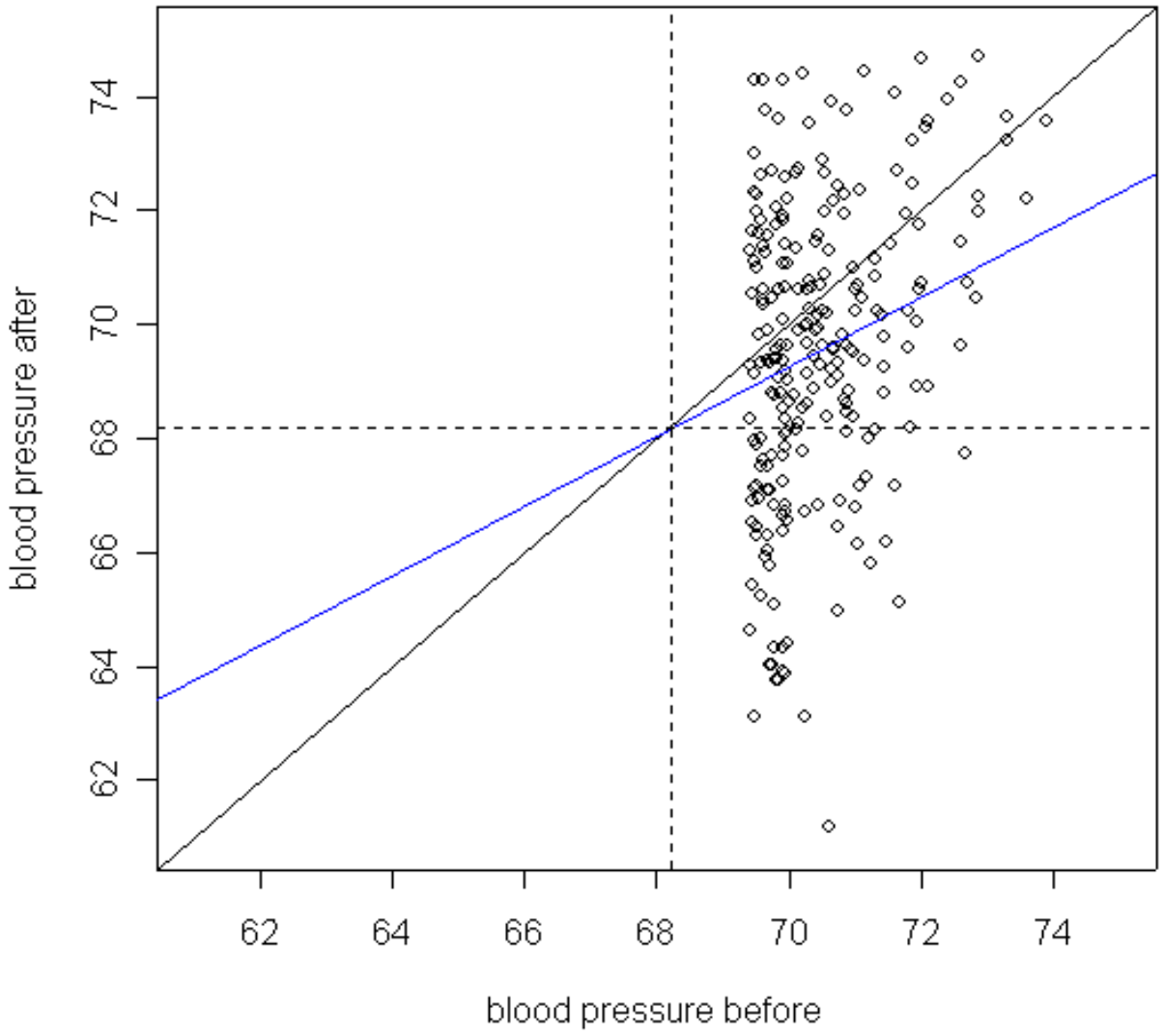
Adjustment for how far parent is from mean of parents

Regression to the Mean: Galton's Height Data



Data: 952 parent-child pairs of heights. Parent height is average of the two parents. Women's heights have been adjusted to make them comparable to men's.

Regression to the Mean is a Powerful Effect!

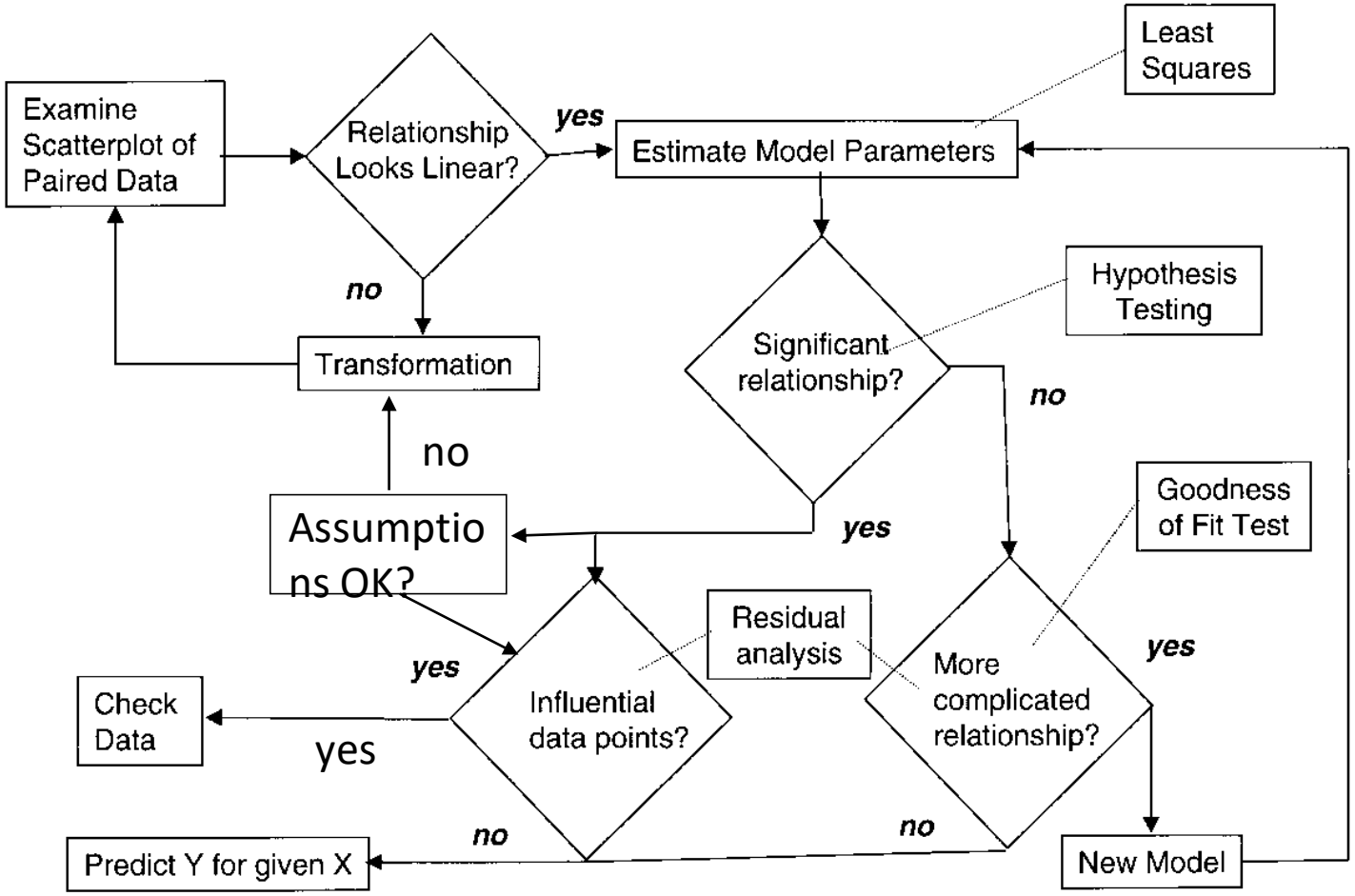


Same data, but suppose response is now blood pressure (bp) before & after (day 1, day 2).

If we track only those with elevated bp before (above 3rd quartile) , we see an amazing improvement, even though no treatment took place!

This is the regression effect at work. If it is not recognized and taken into account, misleading results and biases can occur.

How is a Simple Linear Regression Analysis done? A Protocol



11-16

Steps in a Regression Analysis

1. **Examine the scatterplot of the data.**
   - **Does the relationship look linear?**
   - **Are there points in locations they shouldn't be?**
   - **Do we need a transformation?**
2. **Assuming a linear function looks appropriate, estimate the regression parameters.**
   - **How do we do this? (Method of Least Squares)**
3. **Test whether there really is a statistically significant linear relationship. Just because we assumed a linear function it does not follow that the data support this assumption.**
   - **How do we test this? (F-test for Variances)**
4. **If there is a significant linear relationship, estimate the response, Y, for the given values of X, and compute the residuals.**
5. **Examine the residuals for systematic inadequacies in the linear model as fit to the data.**
   - **Is there evidence that a more complicated relationship (say a polynomial) should be considered; are there problems with the regression assumptions? (Residual analysis).**
   - **Are there specific data points which do not seem to follow the proposed relationship? (Examined using influence measures).**

Simple Linear Regression - Example and Theory

SITUATION: A company that repairs small computers needs to develop a better way of providing customers typical repair cost estimates. To begin this process, they compiled data on repair times (in minutes) and the number of components needing repair or replacement from the previous week. The data, sorted by number of components are as follows:

i	Number of components x_i	Repair time y_i
1	1	23
2	2	29
3	4	64
4	4	72
5	4	80
6	5	87
7	6	96
8	6	105
9	8	127
10	8	119
11	9	145
12	9	149
13	10	165
14	10	154

Paired Observations (x_i, y_i)

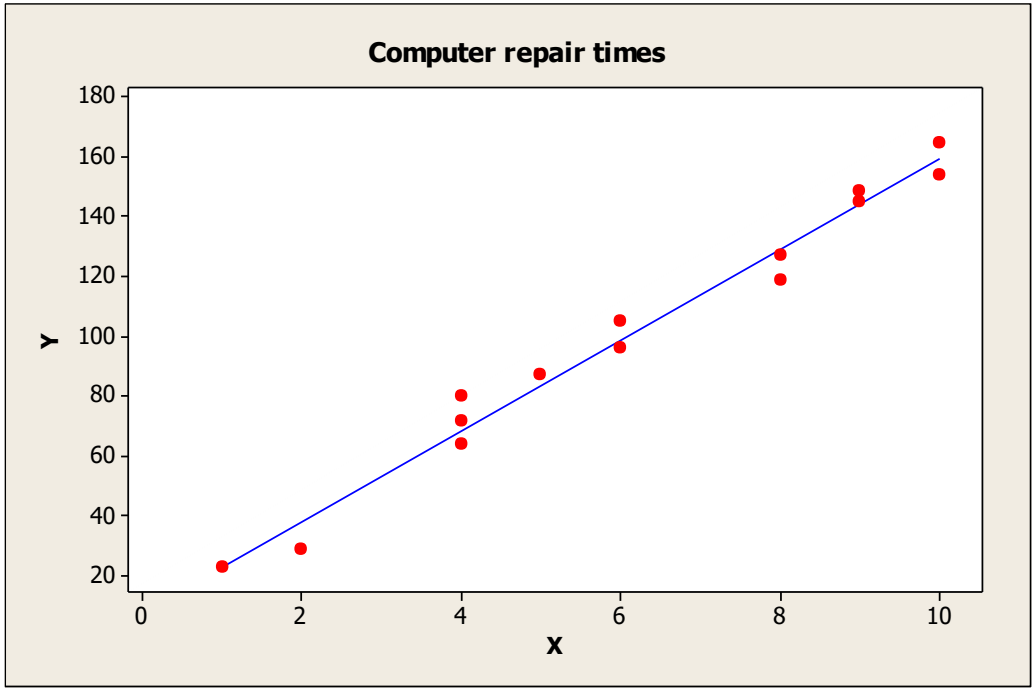
Assumed Linear Regression Model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\text{for } i = 1, 2, \ldots, n$$

Estimating the regression parameters

Objective: Minimize the difference between the observation and its prediction according to the line.



$$\varepsilon_i = y_i - \hat{y}_i$$

$$= y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

$$\hat{y}_i = \text{predicted y value when x} = x_i$$

We want the line which is best for all points. This is done by finding the values of β_0 and β_1 which minimizes some sum of errors. There are a number of ways of doing this. Consider these two:

$$\min_{\beta_0,\beta_1} \sum_{i=1}^{n} |\varepsilon_i|$$

$$\min_{\beta_0,\beta_1} \sum_{i=1}^{n} \varepsilon_i^2 \quad \longleftarrow$$

Sum of squared residuals

The method of least squares produces estimates with statistical properties (e.g. sampling distributions) which are easier to determine.

Regression => least squares estimation

$\hat{\beta}_0 \quad \hat{\beta}_1$ Referred to as least squares estimates.

Normal Equations

Calculus is used to find the least squares estimates.

$$E(\beta_0, \beta_1) = \sum_{i=1}^{n} \varepsilon_i^{\,2} = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\frac{\partial E}{\partial \beta_0} = 0$$

$$\frac{\partial E}{\partial \beta_1} = 0$$

Solve this system of two equations in two unknowns.

Note: The parameter estimates will be functions of the data, hence they will be statistics.

Sums of Squares

Let:

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$= (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2$$

$$= \sum_{i=1}^{n}(x_i^{\,2}) - \frac{1}{n}\left(\sum_{i-1}^{n} x_i\right)^2$$

Sums of squares of x.

$$S_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

$$= (y_1 - \bar{y})^2 + (y_2 - \bar{y})^2 + \cdots + (y_n - \bar{y})^2$$

$$= \sum_{i=1}^{n}(y_i^{\,2}) - \frac{1}{n}\left(\sum_{i-1}^{n} y_i\right)^2$$

Sums of squares of y.

$$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

$$= (x_1 - \bar{x})(y_1 - \bar{y}) + \cdots + (x_n - \bar{x})(y_i - \bar{y})$$

$$= \sum_{i=1}^{n}(x_i y_i) - \frac{1}{n}\left(\sum_{i-1}^{n} x_i\right)\left(\sum_{i-1}^{n} y_i\right)$$

Sums of cross products of x and y.

Parameter estimates:

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Easy to compute with a spreadsheet program.
Easier to do with a statistical analysis package.

Example:

$$\hat{\beta}_1 = 7.71$$

$$\hat{\beta}_0 = 15.20$$

$$\hat{y}_i = 15.20 + 7.71 x_i$$ Prediction

Testing for a Statistically Significant Regression

H_o: There is no relationship between Y and X.

H_A: There is a relationship between Y and X.

Which of two competing models is more appropriate?

$$\text{Linear Model}: \quad Y = \beta_0 + \beta_1 X + \varepsilon$$

$$\text{Mean Model}: \quad Y = \mu + \varepsilon$$

We look at the sums of squares of the prediction errors for the two models and decide if that for the linear model is **significantly smaller** than that for the mean model.

Sums of Squares About the Mean (TSS)

Sum of squares about the mean: sum of the prediction errors for the null (mean model) hypothesis.

$$TSS = S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

TSS is actually a measure of the variance of the responses.

Residual Sums of Squares

Sum of squares for error: sum of the prediction errors for the alternative (linear regression model) hypothesis.

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

SSE measures the variance of the residuals, the part of the response variation that is not explained by the model.

Regression Sums of Squares

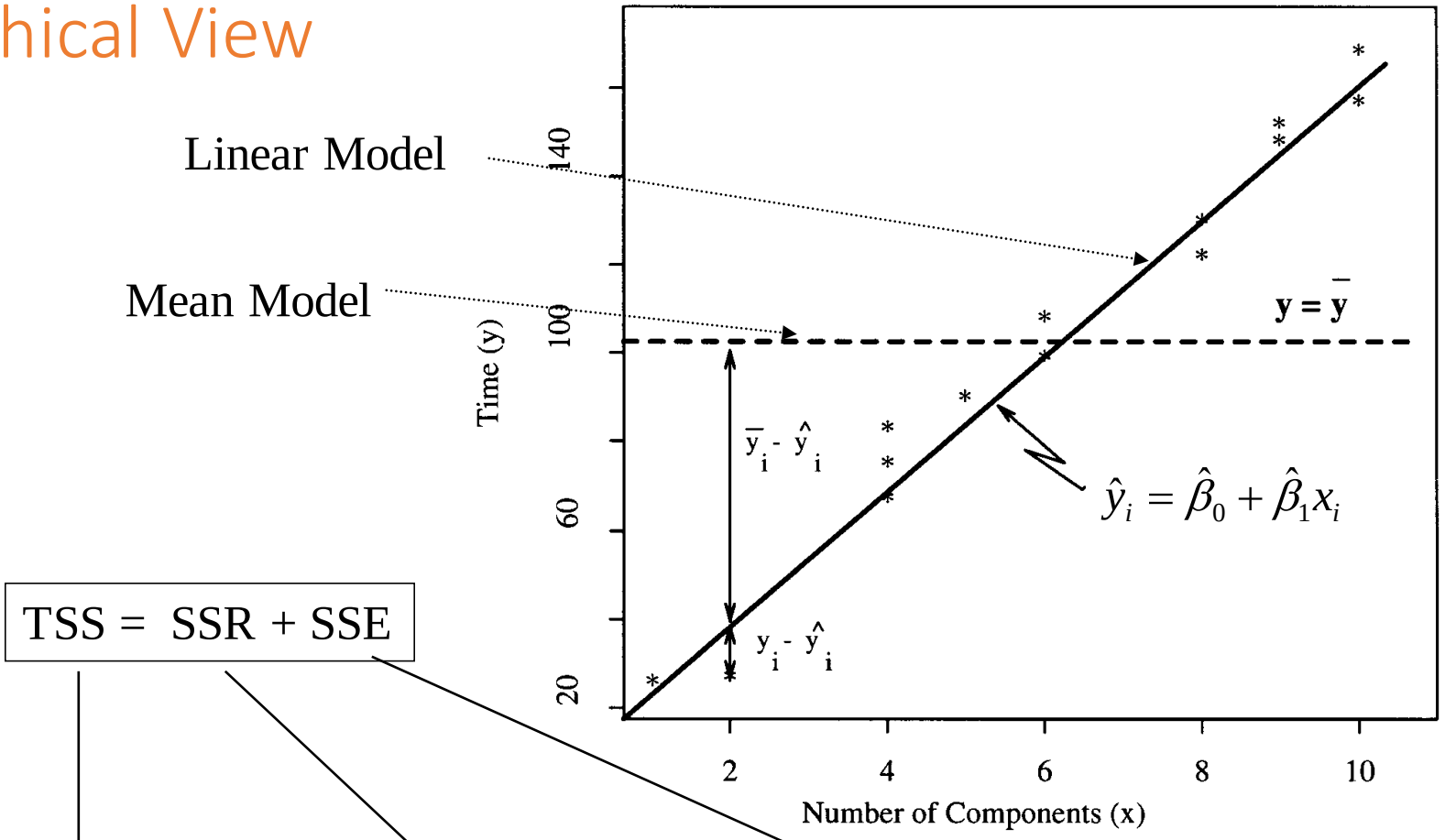
Sum of squares due to the regression: difference between TSS and SSE, i.e. SSR = TSS – SSE.

$$SSR = \sum_{i=1}^{n} (y_i - \bar{y}_i)^2 - \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$= \sum_{i=1}^{n} (\bar{y} - \hat{y}_i)^2$$

SSR measures how much variability in the response is explained by the regression.

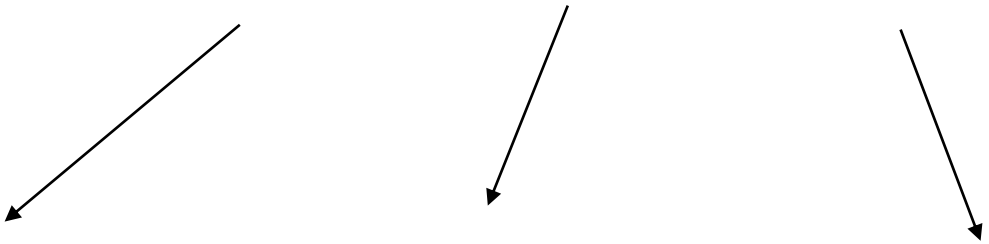
Graphical View



TSS = SSR + SSE

Total variability in y-values = Variability accounted for by the regression + Unexplained variability

$$TSS = SSR + SSE$$



Total variability in y-values $=$ Variability accounted for by the regression $+$ Unexplained variability

regression model fits well

Then SSR approaches TSS and SSE gets small.

regression model adds little

Then SSR approaches 0 and SSE approaches TSS.

Mean Square Terms

Mean Square Total

Sample variance of the response, y:

$$\hat{\sigma}^2_{\mathrm{T}} = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 = \frac{\mathrm{TSS}}{n-1} = \mathrm{MST}$$

$$\hat{\sigma}^2_{\mathrm{R}} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 = \frac{\mathrm{SSR}}{1} = \mathrm{MSR}$$

Regression Mean Square:

Residual Mean Square

$$\hat{\sigma}^2_{\varepsilon} = \hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{\mathrm{SSE}}{n-2} = \mathrm{MSE}$$

F Test for Significant Regression

Both MSE and MSR measure the same underlying variance quantity under the assumption that the null (mean) model holds.

$$\sigma_R^2 \approx \sigma_\varepsilon^2$$

Under the alternative hypothesis, the MSR should be much greater than the MSE.

$$\sigma_R^2 > \sigma_\varepsilon^2$$

Placing this in the context of a test of variance.

$$F = \frac{\sigma_R^2}{\sigma_\varepsilon^2} = \frac{\text{MSR}}{\text{MSE}}$$ Test Statistic

F should be near 1 if the regression is *not significant*, i.e. H_0: mean model holds.

Formal test of the significance of the regression.

H_0: No significant regression fit.

H_A: The regression explains a significant amount of the variability in the response.

or

The slope of the regression line is significant.

or

X is a significant predictor of Y.

Test Statistic: $$F = \frac{MSR}{MSE}$$

Reject H_0 if: $$F > F_{1,n-2,\alpha}$$

Where α is the probability of a type I error.

Assumptions

1. $\varepsilon_1, \varepsilon_2, \ldots \varepsilon_n$ **are independent of each other.**
2. **The** ε_i **are normally distributed with mean zero and have common variance** σ^2**.**

How do we check these assumptions?

I. **Appropriate graphs.**
II. **Correlations (more later).**
III. **Formal goodness of fit tests.**

Analysis of Variance Table

We summarize the computations of this test in a table.

Source	Sums of Squares SSQ	Degrees of Freedom DF	Mean Squares MS	F
Regression	SSR	1	MSR	$F = \frac{MSR}{MSE}$
Error	SSE	n-2	MSE	
Total	SSM	n-1		



TSS

i	Number of components x_i	Repair time y_i
1	1	23
2	2	29
3	4	64
4	4	72
5	4	80
6	5	87
7	6	96
8	6	105
9	8	127
10	8	119
11	9	145
12	9	149
13	10	165
14	10	154

```
*-------------------------------------------------------*;
*   Set up linesize (ls) and pagesize (ps) parameters     *;
*-------------------------------------------------------*;
options ls=78 ps=40 nodate;
data repair;
infile 'repair.txt';
input ncomp time;
label ncomp="No. of components" time="Repair time";
run;
*-------------------------------------------------------*;
*  The regression analysis procedure (PROC REG) is run.   *;
*   We ask for a printout of                             *;
*       predicted values (p),  residual values (r)       *;
*       confidence intervals and prediction intervals    *;
*       for y (cli, clm).  Other additional statistics   *;
*       will also be printed out, including statistics   *;
*       on the influence of observations on the model fit*;
*   We also ask for various plots to be produced to allow  *;
*     examination of model fit and assumptions           *;
*-------------------------------------------------------*;
proc reg ;
 model time = ncomp / p r cli clm influence;
 title 'STA6166 - Regression Example';
 plot time*ncomp p.*ncomp='+'/ overlay symbol='*';
```

SAS output

Table 2: SAS Output

STA6166 - Regression Example 1

Model: MODEL1 (1)
Dependent Variable: TIME Repair time

(2) Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Prob>F
Model	1	25804.38893	25804.38893	623.621	0.0001
Error	12	496.53964	41.37830		
C Total	13	26300.92857			

(3)	Root MSE	6.43260	R-square	0.9811	(4)
(5)	Dep Mean	101.07143	Adj R-sq	0.9795	(6)
(7)	C.V.	6.36441			

MSE

$$\hat{\sigma}_\varepsilon = \sqrt{\text{MSE}}$$

Parameter Standard Error Estimates

Under the assumptions for regression inference, the least squares estimates themselves are random variables.

1. $\varepsilon_1, \varepsilon_2, \ldots \varepsilon_n$ are independent of each other.
2. The ε_i are normally distributed with mean zero and have common variance σ^2.

Using some more calculus and mathematical statistics we can determine the distributions for these parameters.

$$\hat{\beta}_0 \mapsto \mathrm{N}\left(\beta_0, \sigma^2 \frac{\sum x_i^2}{nS_{XX}}\right) \qquad \hat{\beta}_1 \mapsto \mathrm{N}\left(\beta_1, \frac{\sigma^2}{S_{XX}}\right)$$

Testing regression parameters

The estimate of σ^2 is the mean square error: MSE

important ←

$$\hat{\sigma}^2 = MSE$$

Test H_0: $\beta_1=0$:

$$t_{\beta_1} = \frac{\hat{\beta}_1 - 0}{\sqrt{MSE / S_{XX}}}$$

Reject H_0 if:

$$\left| t_{\beta_1} \right| > t_{n-2,\alpha/2}$$

(1-α)100% CI for β_1:

$$\hat{\beta}_1 \pm t_{n-2,\alpha/2} \sqrt{\frac{MSE}{S_{XX}}}$$

$\hat{\beta}_0$

P-values

Table 3: SAS Output

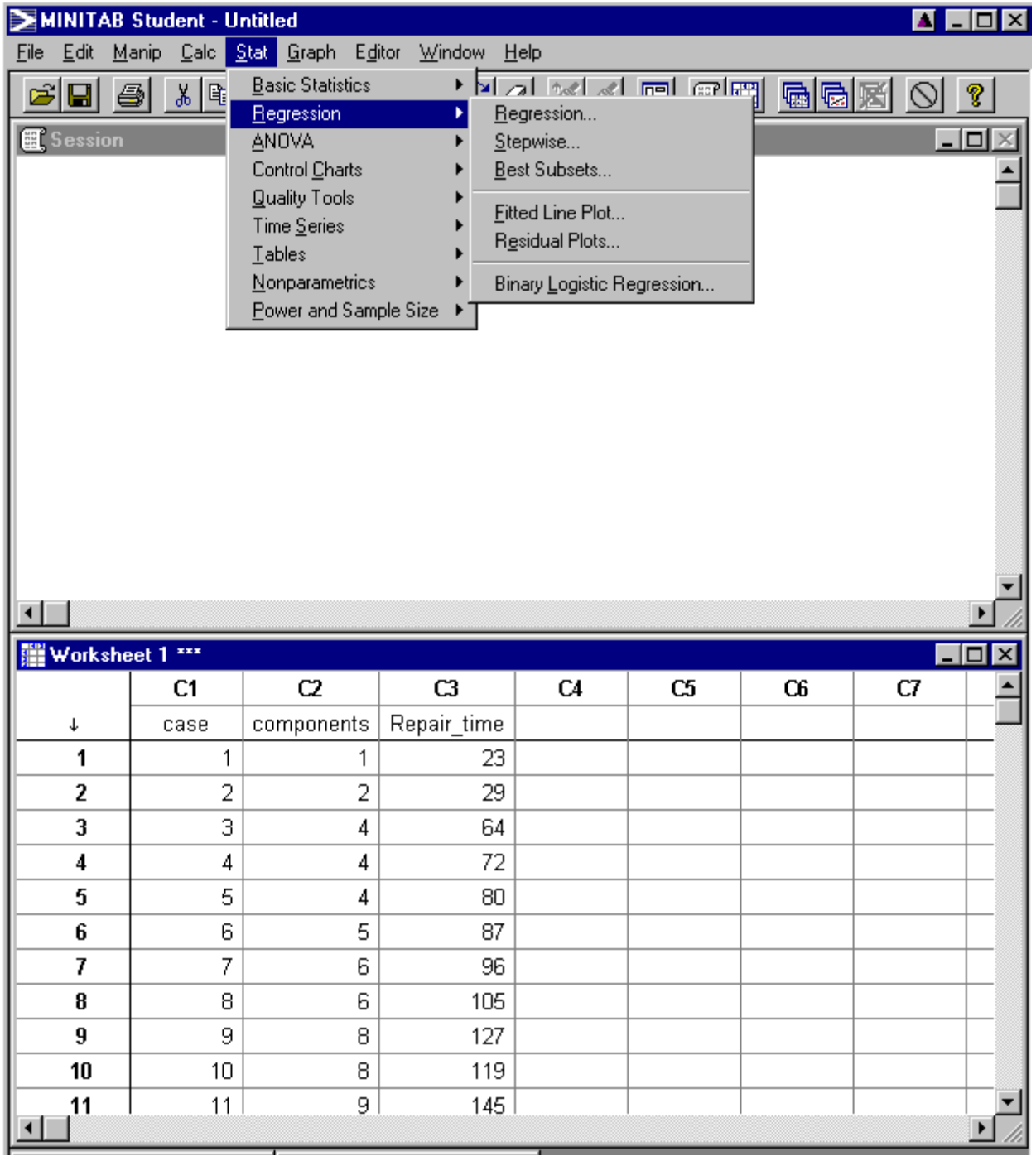
Parameter Estimates

Variable	DF	(1) Parameter Estimate	(2) Standard Error	(3) T for H0: Parameter=0	(4) Prob > |T|
INTERCEP	1	7.710997	4.11488737	1.874	0.0855
NCOMP	1	15.198210	0.60860005	24.972	0.0001

$\hat{\beta}_1$

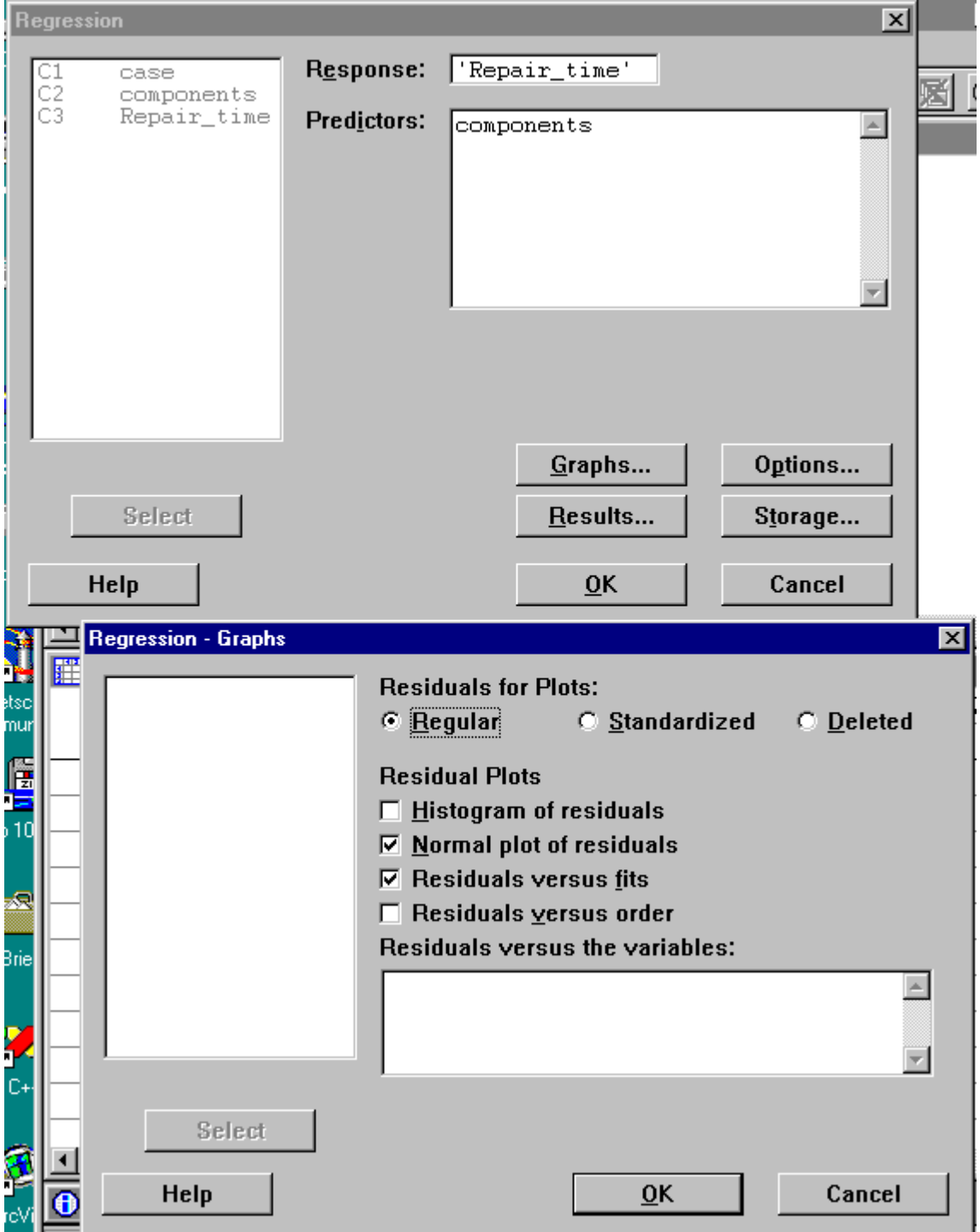
$\sqrt{MSE/S_{XX}}$

$$t_{\beta_1} = \frac{\hat{\beta}_1 - 0}{\sqrt{MSE/S_{XX}}}$$



	C1	C2	C3	C4	C5	C6	C7
↓	case	components	Repair_time				
1	1	1	23				
2	2	2	29				
3	3	4	64				
4	4	4	72				
5	5	4	80				
6	6	5	87				
7	7	6	96				
8	8	6	105				
9	9	8	127				
10	10	8	119				
11	11	9	145				

Regression in Minitab



Specifying Model and Output Options

Session

Regression Analysis

The regression equation is
Repair_time = 7.71 + 15.2 components

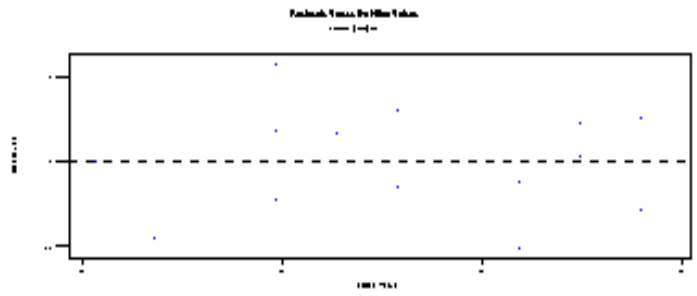
Predictor	Coef	StDev	T	P
Constant	7.711	4.115	1.87	0.085
componen	15.1982	0.6086	24.97	0.000

S = 6.433 R-Sq = 98.1% R-Sq(adj) = 98.0%

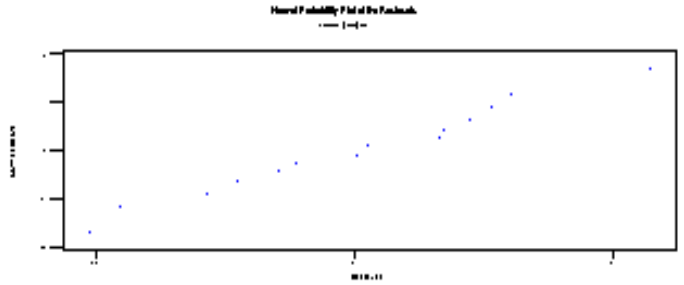
Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	25804	25804	623.62	0.000
Residual Error	12	497	41		

Worksheet 1: Res vs Fits for Repair_t



Worksheet 1: Normplot of Res for Repair_t



Regression in R

```
> y_c(23,29,64,72,80,87,96,105,127,119,145,149,165,154)
> x_c(1,2,4,4,4,5,6,6,8,8,9,9,10,10)
> myfit <- lm(y ~ x)
> summary(myfit)

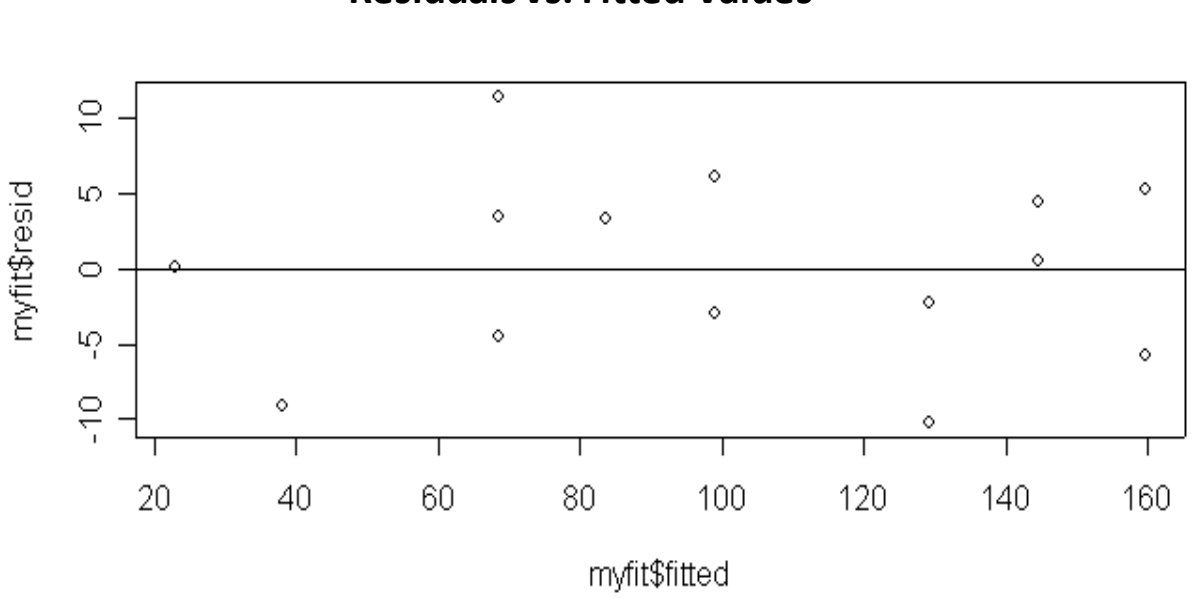
Residuals:
    Min      1Q  Median      3Q     Max
-10.2967  -4.1029   0.2980   4.2529  11.4962

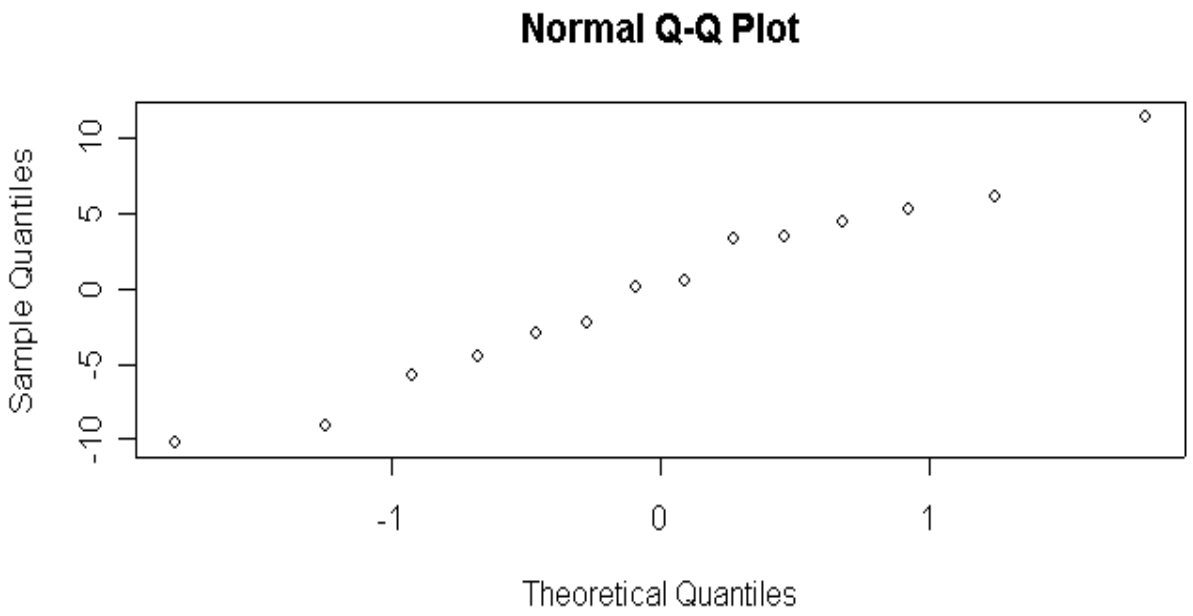
Coefficients:
            Estimate   Std. Error  t value  Pr(>|t|)
(Intercept)   7.7110      4.1149    1.874    0.0855 .
x            15.1982      0.6086   24.972   1.03e-11 ***
---
Signif. codes:  0 `***' 0.001 `**' 0.01 `*' 0.05 `.' 0.1 ` ' 1

Residual standard error: 6.433 on 12 degrees of freedom
Multiple R-Squared: 0.9811,     Adjusted R-squared: 0.9795
F-statistic: 623.6 on 1 and 12 DF,  p-value: 1.030e-11

> anova(myfit)
Analysis of Variance Table

Response: y
```






```
> par(mfrow=c(2,1))
>
plot(myfit$fitted,myfit$resi
d)
> abline(0,0)

> qqnorm(myfit$resid)
```